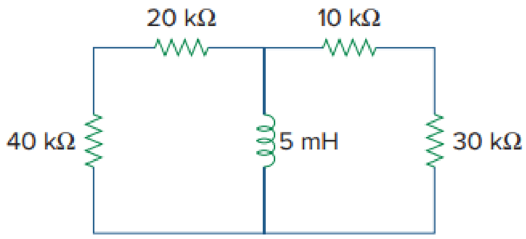
Chapter 7, Problem 14P

Problem

Calculate the time constant of the circuit in Fig. 7.94.

Fig. 7.94:



Step-by-step solution

Step 1 of 5

Consider the following circuit diagram:

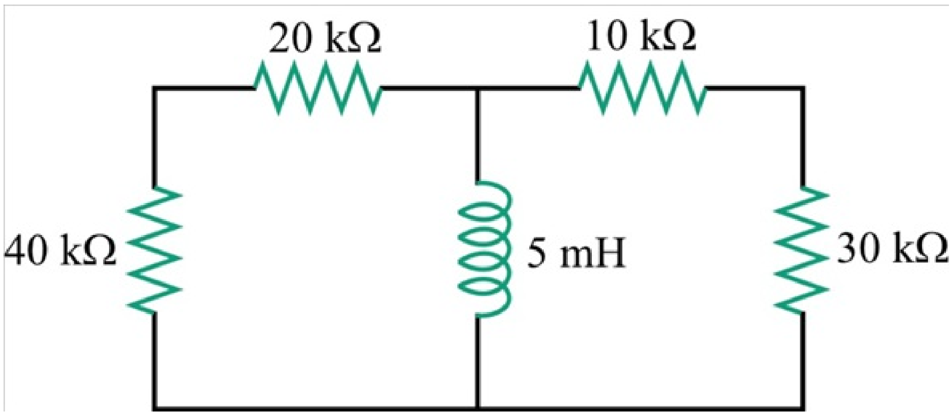


Figure 1

Step 2 of 5

Write the expression for time constant for LC circuit.

$$\tau = \frac{L}{R_{eq}}$$

Where,

L is the inductance

R_{eq} is the equivalent resistance

Step 3 of 5

Remove the inductor from the circuit shown in Figure 2, draw the modified circuit diagram:

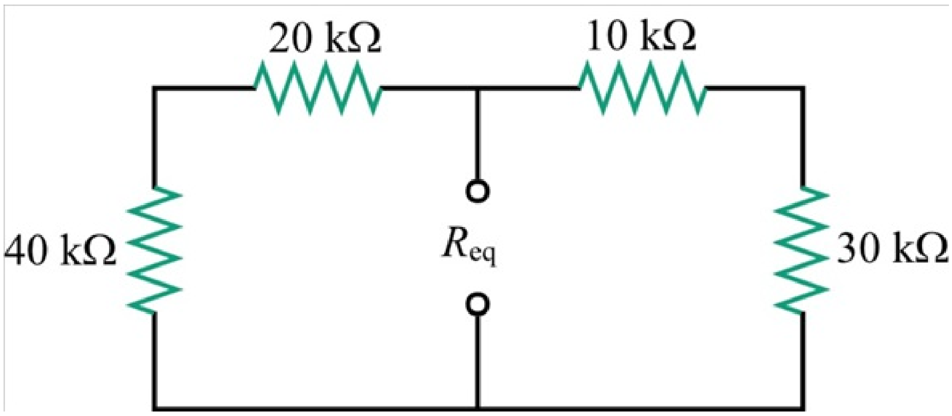


Figure 2

Step 4 of 5

Looking from the open terminals of the inductor, series connected $40\ \text{k}\Omega$ and $20\ \text{k}\Omega$ resistors are in parallel with series connected $40\ \text{k}\Omega$ and $20\ \text{k}\Omega$ resistors. Calculate the equivalent resistance.

$$\begin{aligned} R_{eq} &= (40\ \text{k}\Omega + 20\ \text{k}\Omega) \,||\, (10\ \text{k}\Omega + 30\ \text{k}\Omega) \\ &= (60\ \text{k}\Omega) \,||\, (40\ \text{k}\Omega) \\ &= \frac{(60\ \text{k}\Omega)(40\ \text{k}\Omega)}{60\ \text{k}\Omega + 40\ \text{k}\Omega} \\ &= \frac{2400\ \text{k}\Omega}{100} \end{aligned}$$

$$= 24\ \text{k}\Omega$$

Step 5 of 5

Now calculate the time constant.

$$\begin{aligned} \tau &= \frac{L}{R_{eq}} \\ &= \frac{5\ \text{mH}}{24\ \text{k}\Omega} \\ &= 0.20833\ \mu\text{s} \\ &= 208.33\ \text{ns} \end{aligned}$$

Therefore, the time constant, τ is $\boxed{208.33\ \text{ns}}$.